Homework 2
Statistics 2103

1. Let V be a random variable representing a student's score on the verbal section of the GRE exam. Suppose the distribution of V has the following characteristics:

$$\mathbb{E}(V) = 151, \quad \text{var}(V) = 8^2$$

The file `scores.csv` contains the math and verbal scores for 1000 randomly selected students.

   (a) Make a histogram of the verbal scores, and on top of the histogram, plot the probability density function of a normal distribution with mean 151 and standard deviation 8.

   (b) Simulate 3000 samples from a normal distribution with mean 151 and standard deviation 8. What fraction of the sample values were less than 130? What fraction were greater than 170?

   (c) In reality, each section of the GRE exam only gives scores in the 130-170 range. Based on your results in parts (a) and (b), do you think GRE verbal section scores follow a Normal$(151, 8^2)$ distribution? If not, do you think the Normal distribution still provides a good approximation to the distribution of the scores? Explain in a few sentences.

2. Suppose that 16% of children in Alaska are homeschooled.

   (a) If you randomly select 4 children in Alaska, what is the probability that at least one of them is homeschooled?

   (b) If you randomly select 30 children in Alaska, what is the probability that at least 20% of them are homeschooled?

   (c) What is the smallest number of children would you need to sample to ensure that the probability of getting two or more homeschooled children in your sample is at least 75%?

3. Suppose you have a bird feeder in your backyard, and the amount of birdseed consumed by the birds in one day is normally distributed with a mean of 400 grams and standard deviation of 20 grams.

(a) How much birdseed should you put in the feeder to ensure that the probability of running out by the end of the day is at most 5%?

(b) If the amounts consumed each day are independent, how much birdseed should you buy for a 7-day week to ensure that the probability of running out by the end of the week is at most 5%?

4. Suppose the probability that a sea turtle hatchling is born female is p. Scientists are interested in whether this probability has changed over time due to changes in ocean temperatures. Scientists observe a random sample of 90 sea turtle hatchlings, of which 56 are born female.

(a) If the true probability p were 0.5, what number of female sea turtle hatchlings would you expect to see in a random sample of 90?

(b) If the true probability p were 0.5, what is the probability that in a sample of 90, at least 56 would be female?

(c) Based on the sample data, give a 95% confidence interval for the probability that a sea turtle hatchling is born female.

(d) Based on your answers to parts (a) - (c), do you think the true probability is actually 0.5? Why or why not?